

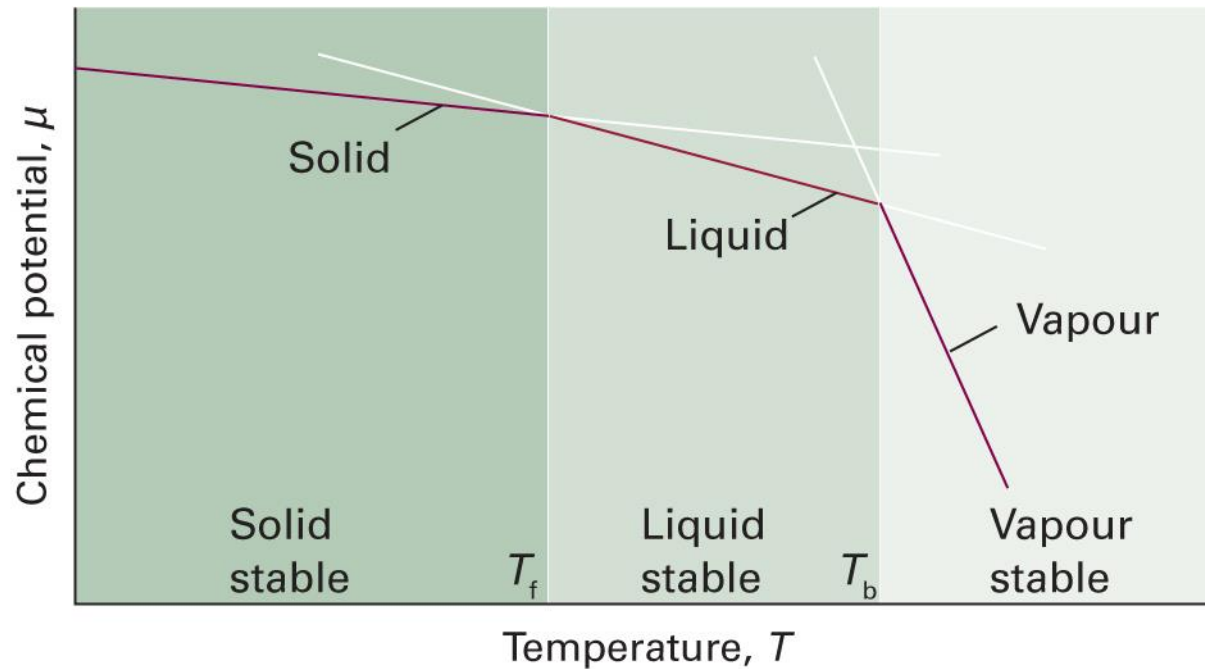
Physical Chemistry of Materials

Lecture 04

Foundations of Statistical Mechanics



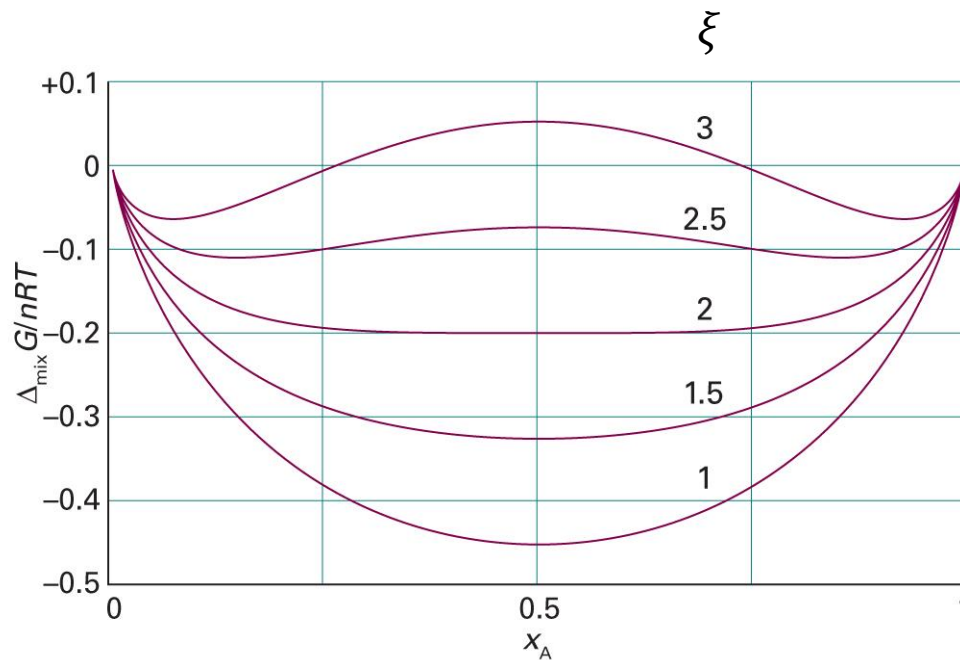
Chemical Potential



(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figures 4B.1)

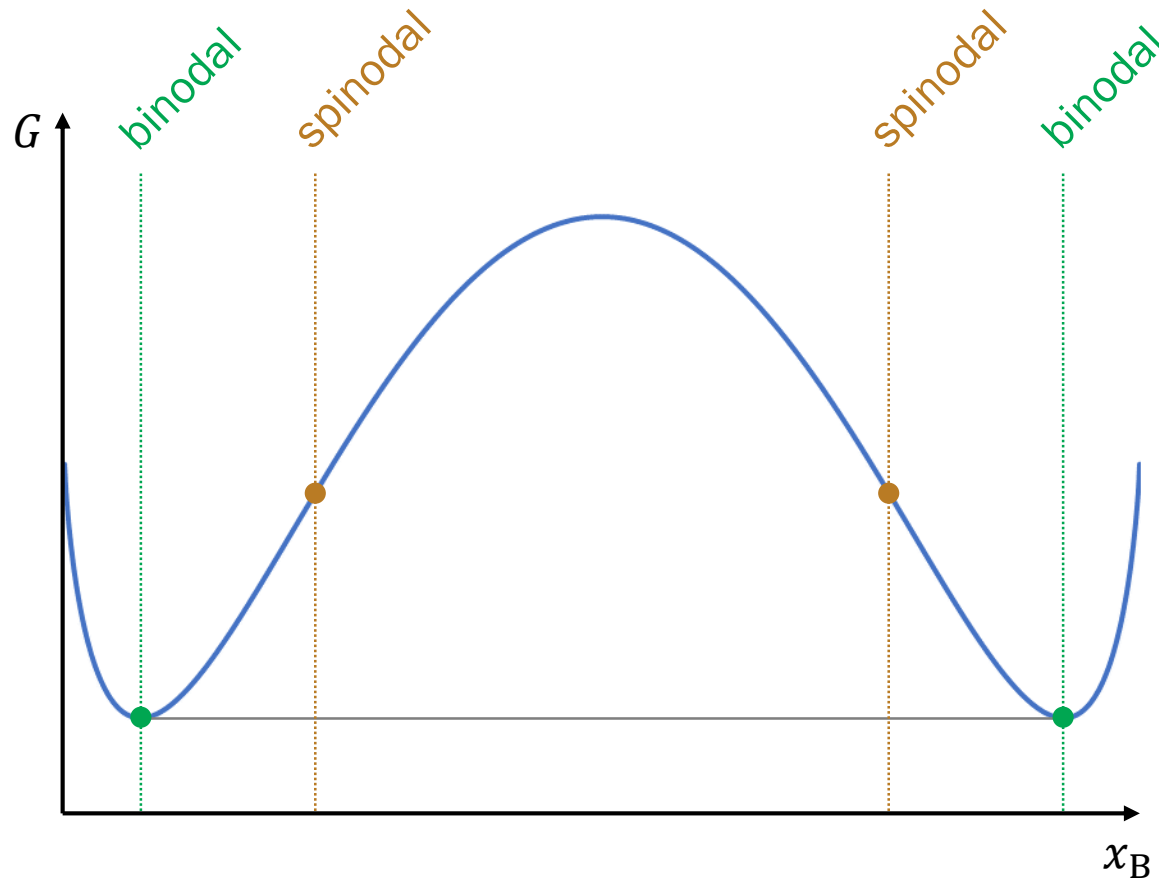
Gibbs Energy of Mixing

$$\Delta_{\text{mix}}G = nRT(x_A \ln x_A + x_B \ln x_B + \xi x_A x_B)$$



(Image: Atkins et al., *Atkins' Physical Chemistry*, 12th ed., Figure 5B.5)

Binodal and Spinodal



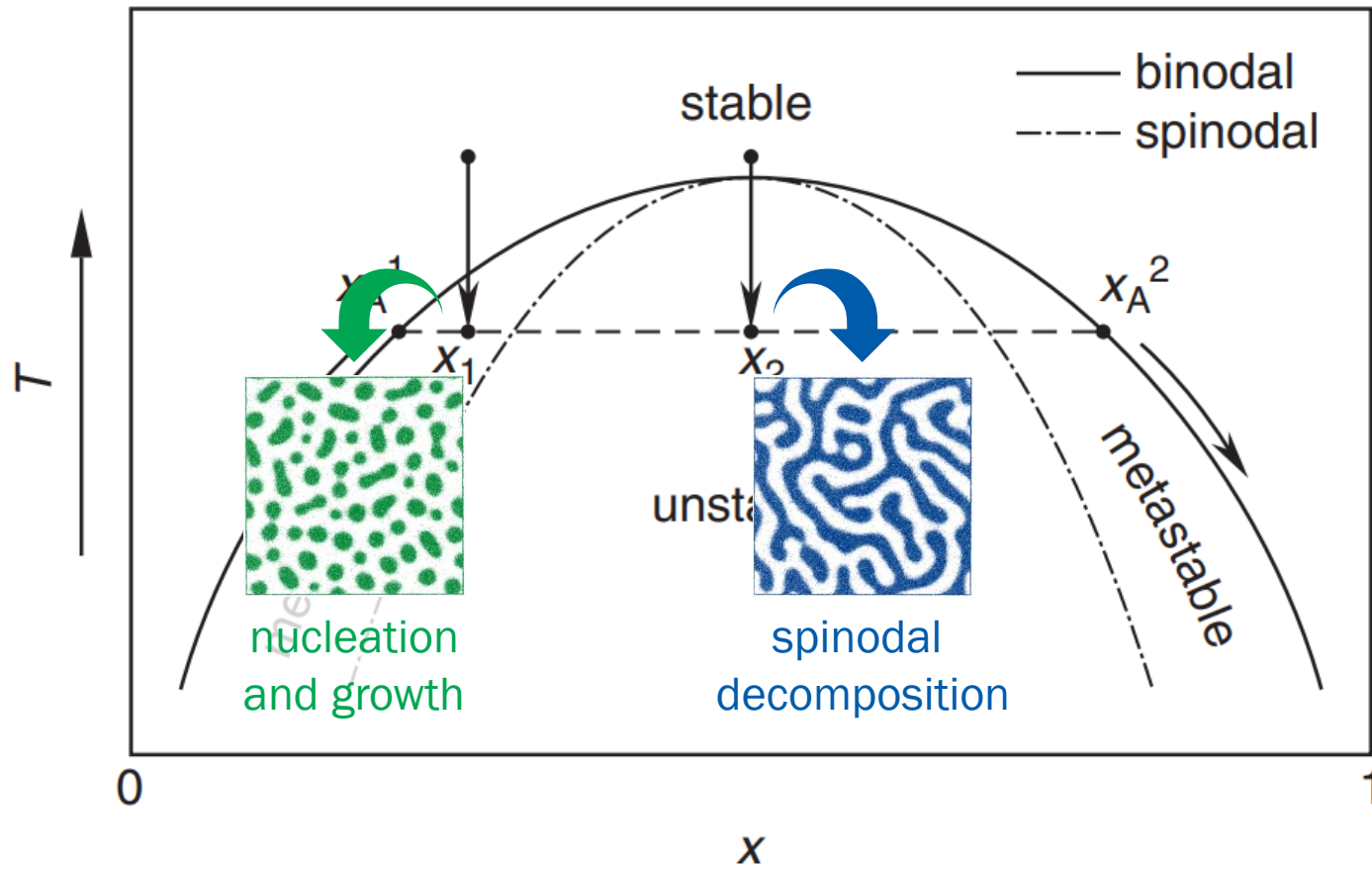
Binodal:

$$\frac{\partial G}{\partial x_B} = 0$$

Spinodal:

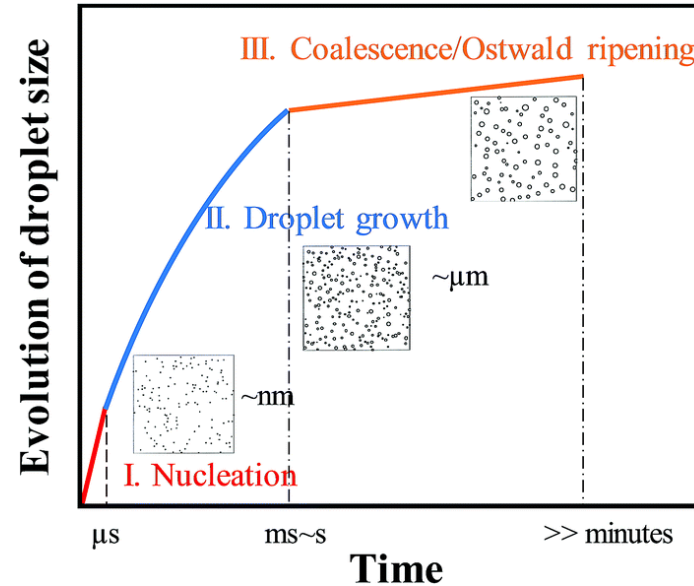
$$\frac{\partial^2 G}{\partial x_B^2} = 0$$

Kinetics of Phase Separation



Nucleation and Growth

metastable phase $\rightarrow$ stable phase



Universal in chemistry and physics!

(Image: Luo et al., *Phys Chem Chem Phys* (2017), Fig. 6)

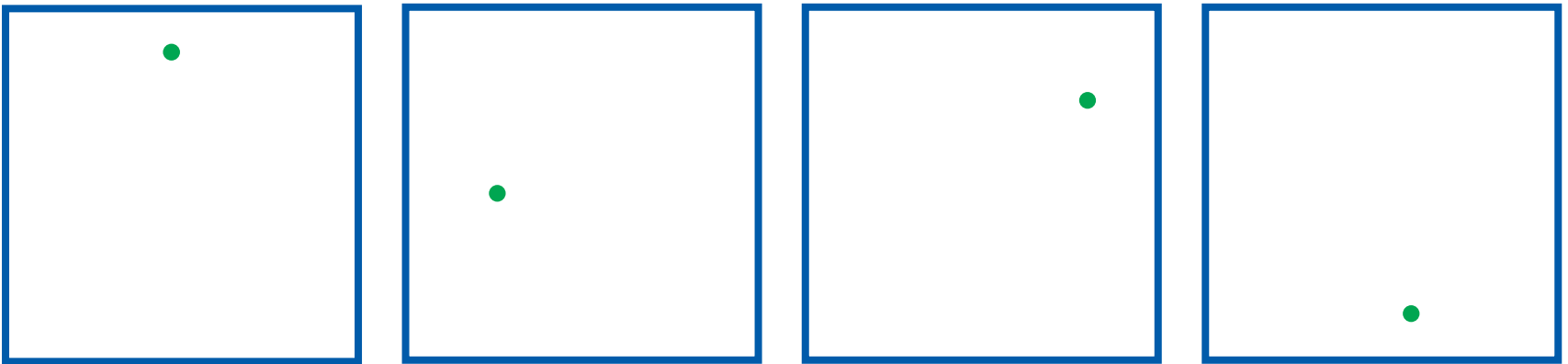
“States”

- (Macro)state: defined by values of **macroscopic** variables
- Microstate: specific **microscopic** arrangement
- Different microstates can be mapped into the same macrostate

Example: Diffusion

When a gas molecule moves around the system,

- **Macrostate:** defined by the volume V
- **Microstate:** the position of the molecule



(Thermodynamic) Ensemble

Idealized group of a large number of virtual copies
(sometimes infinitely many) of a system

Microcanonical ensemble

Canonical ensemble

Grand canonical ensemble

⋮

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Microcanonical Ensemble

Thermodynamic ensemble of an **isolated system**

- Conserved quantities: N, V, U (or traditionally N, V, E)

Fundamental Assumption of SM

“In an isolated system in equilibrium,
all accessible microstates are **equally probable**.”

Microcanonical Ensemble

Thermodynamic ensemble of an **isolated system**

- Conserved quantities: N, V, U (or traditionally N, V, E)
- All microstates are equally probable
- Probability to find the system in microstate j

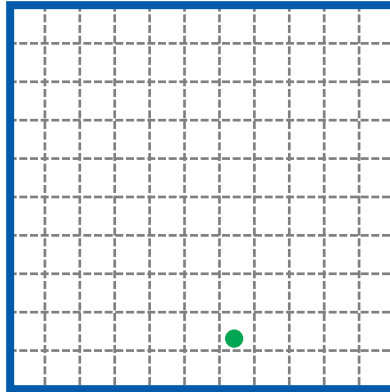
$$p_j = \frac{1}{W}$$

Statistical Definition of Entropy

- W = number of microstates belonging to a specific macrostate (multiplicity, degeneracy)
- Statistical definition of entropy (for an isolated system in equilibrium)

$$S = k_B \ln W$$

Example: Diffusion



Introduce the “unit volume” v_0 .

For the system volume V , $W(V) = V/v_0$

Example: Diffusion

Suppose that the system volume changes from V_i to V_f :

$$W_i = V_i/v_0$$

$$W_f = V_f/v_0$$

The change in entropy is given by

$$\Delta S_1 = k_B \ln(W_f/W_i) = k_B \ln(V_f/V_i)$$

For an ideal gas consisting of N molecules,

$$\Delta S_N = Nk_B \ln(V_f/V_i) = nR \ln(V_f/V_i)$$

(Thermodynamic) Ensemble

Idealized group of a large number of virtual copies
(sometimes infinitely many) of a system

Microcanonical ensemble

Canonical ensemble

Grand canonical ensemble

⋮

Canonical Ensemble

Thermodynamic ensemble of a closed system

- Conserved quantities: N, V, T
- Microstates may have different energies
- All microstates are NOT equally probable
- Probability to find the system in some microstate?

Shannon Entropy

Consider a system with the following properties:

- t microstates
- each microstate i with probability p_i
- Ensemble: $\mathcal{N}$ independent replicates ($\mathcal{N} \rightarrow \infty$)

Among $\mathcal{N}$, the number of finding microstate i is $n_i = \mathcal{N}p_i$

Then, the number of combinations is

$$W = \frac{\mathcal{N}!}{n_1! n_2! \cdots n_t!}$$

Stirling's Approximation

$$\ln N! \approx N \ln N - N \text{ (for large } N\text{)}$$

$$\begin{aligned}\ln N! &= \ln\{N \times (N - 1) \times (N - 2) \times \cdots\} \\&= \ln N + \ln(N - 1) + \ln(N - 2) + \cdots \\&= \sum_{k=1}^N \ln k \\&\approx \int_1^N \ln x \, dx \\&= [x \ln x - x]_1^N \\&= N \ln N - N + 1 \\&\approx N \ln N - N \text{ (for large } N \gg 1\text{)}\end{aligned}$$

Shannon Entropy

$$W = \frac{\mathcal{N}!}{n_1! n_2! \cdots n_t!}$$

$$S = k_B \ln W = k_B \ln \frac{\mathcal{N}!}{n_1! n_2! \cdots n_t!}$$

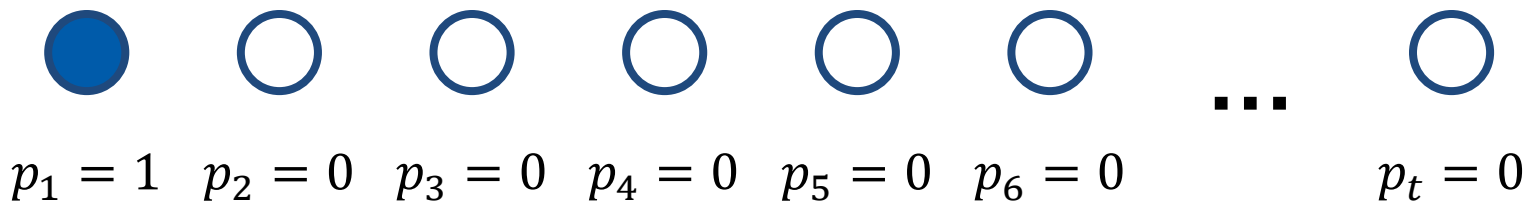
By Stirling's approximation,

$$S \approx k_B (\mathcal{N} \ln \mathcal{N} - \sum_i n_i \ln n_i) = -k_B \sum_i n_i \ln(n_i/\mathcal{N})$$

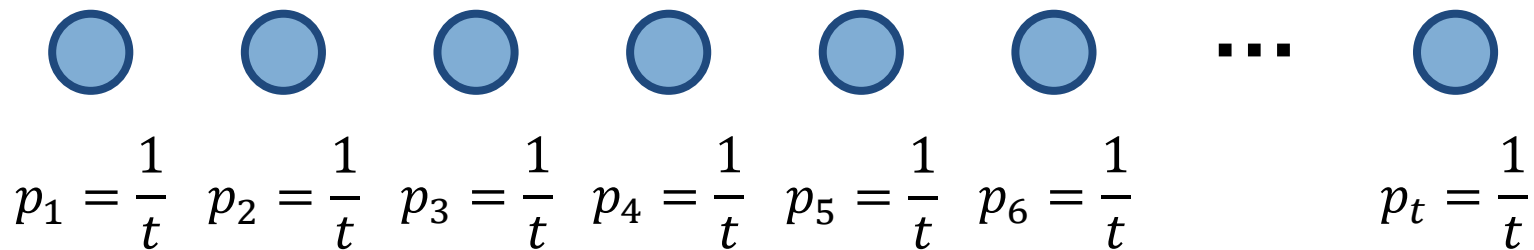
$$S = -\mathcal{N} k_B \sum_i p_i \ln p_i$$

Hence, “average” entropy per system is $S = -k_B \sum_i p_i \ln p_i$

Entropy and Information



$S = 0$ (minimum)



$S = \text{maximum}$

Principle of Maximum Entropy

The probability distribution which **best represents** the current state of knowledge about a system is **the one with largest entropy**

Maximize Shannon Entropy

$$S/k_B = -\sum_i p_i \ln p_i$$

We always have the normalization constraint, $\sum_i p_i - 1 = 0$.

Apply Lagrange's multiplier method:

$$\frac{\partial [S/k_B - \lambda(\sum_i p_i - 1)]}{\partial p_i} = -(\ln p_i + 1) - \lambda = 0$$

$$p_i = e^{-1-\lambda}$$

Using the normalization constraint, $p_i = 1/W$

(microcanonical ensemble)



Ensemble Energy Conservation

$$S/k_B = -\sum_i p_i \ln p_i$$

$$\sum_i p_i - 1 = 0$$

We can introduce one more constraint: $\sum_i n_i E_i = \mathcal{N}U$

As the constraint implies $\sum_i p_i E_i - U = 0$,

$$\frac{\partial [S/k_B - \lambda(\sum_i p_i - 1) - \beta(\sum_i p_i E_i - U)]}{\partial p_i} = 0$$

$$p_i = e^{-1-\lambda-\beta E_i}$$

Partition Function

$$p_i = e^{-1-\lambda-\beta E_i}$$

From the normalization constraint,

$$\sum_i p_i = e^{-1-\lambda} \sum_i e^{-\beta E_i} = 1$$

Define the **partition function Z** :

$$Z = e^{1+\lambda} = \sum_i \frac{e^{-\beta E_i}}{\text{Boltzmann factor}}$$

Back to Entropy

$$p_i = e^{-\beta E_i} / Z$$

$$S/k_B = - \sum_i p_i \ln p_i = - \sum_i p_i \ln \left(\frac{e^{-\beta E_i}}{Z} \right)$$

$$S/k_B = - \sum_i p_i (-\beta E_i - \ln Z) = \beta \sum_i p_i E_i + \ln Z$$

$$\therefore S/k_B = \beta U + \ln Z$$

Bridge to Thermodynamics

$$S = k_B \beta U + k_B \ln Z$$

From thermodynamics, we know

$$\left(\frac{\partial S}{\partial U} \right)_{V,n} = \frac{1}{T}$$

Hence, $k_B \beta = 1/T$ or $\beta = 1/k_B T$ (inverse temperature)

Also,

$$-k_B T \ln Z = U - TS = A \text{ (Helmholtz energy)}$$

Summary

For a canonical ensemble, the probability to find the system in a microstate i (with energy E_i) is

$$p_i = e^{-1-\lambda-\beta E_i} = \frac{e^{-\beta E_i}}{Z} = \frac{e^{-E_i/k_B T}}{Z}$$

where the partition function $Z = \sum_i e^{-\beta E_i} = \sum_i e^{-E_i/k_B T}$

Also, the Helmholtz energy is given by

$$A = -k_B T \ln Z$$

Derived Functions

$$A = -k_{\text{B}}T \ln Z$$

Using $dA = -SdT - pdV + \mu dn$,

$$S = -\left(\frac{\partial A}{\partial T}\right)_{V,n}, \quad p = -\left(\frac{\partial A}{\partial V}\right)_{T,n}, \quad \mu = \left(\frac{\partial A}{\partial n}\right)_{T,V}$$

One More Useful Relation

$$U = \sum_i p_i E_i = \sum_i \frac{E_i e^{-\beta E_i}}{Z}$$

However,

$$\frac{\partial \ln Z}{\partial \beta} = \frac{1}{Z} \frac{\partial Z}{\partial \beta} = -\frac{1}{Z} \sum_i E_i e^{-\beta E_i}$$

Hence,

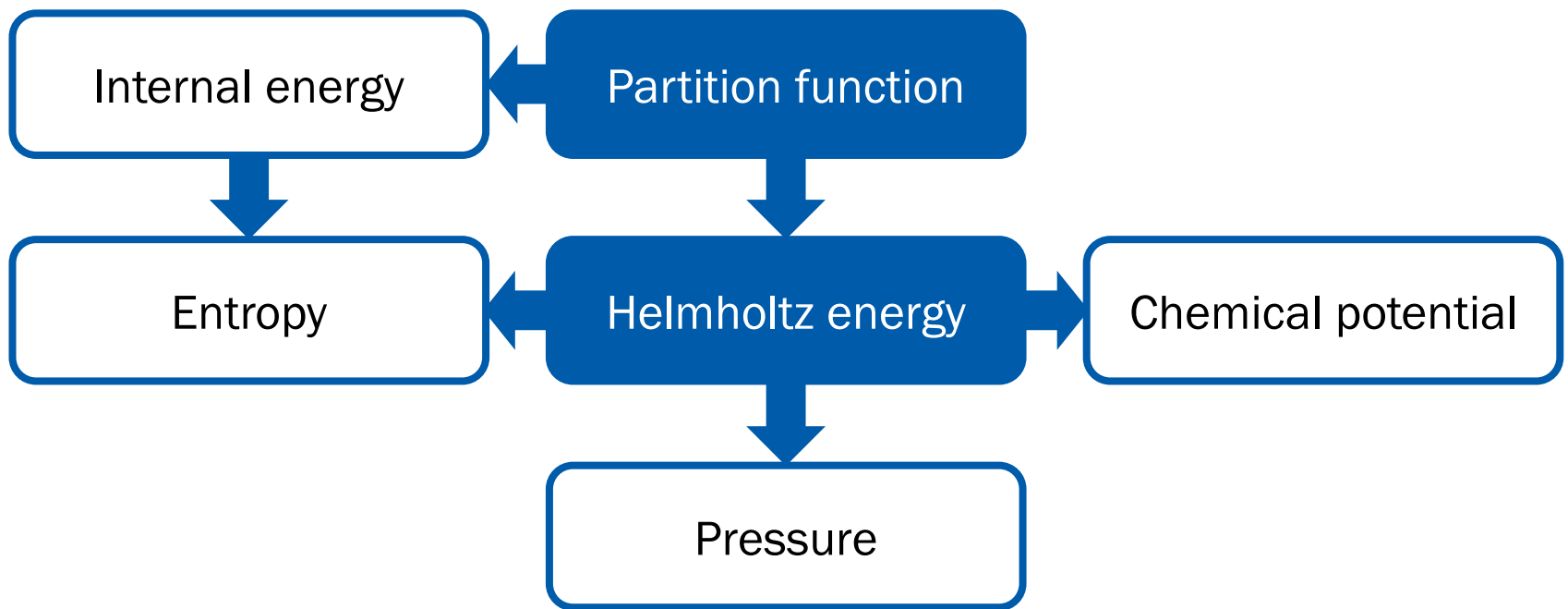
$$U = -\frac{\partial \ln Z}{\partial \beta}$$

Partition Function and ...

- Helmholtz energy: $A = -k_{\text{B}}T \ln Z$
- Internal energy: $U = -\partial(\ln Z)/\partial\beta$
- Entropy: $S = (U - A)/T$

“A partition function contains
all the **thermodynamic information** about a system.”

Z and Thermodynamics



Multi-Particle Partition Functions

- Independent & distinguishable particles

$$Z = z^N \text{ (} z: \text{ one-particle partition function)}$$

- Independent & indistinguishable particles

$$Z = \frac{1}{N!} z^N$$

- Interacting particles: later

Two Distinguishable Particles

- Consider two particles A and B, with energies ε_i^A and ε_j^B
- Partition function of A: $z_A = \sum_i e^{-\beta \varepsilon_i^A}$
- Partition function of B: $z_B = \sum_j e^{-\beta \varepsilon_j^B}$
- Total energy: $E_{ij} = \varepsilon_i^A + \varepsilon_j^B$
- Total partition function

$$Z = \sum_{i,j} e^{-\beta E_{ij}} = \sum_i \sum_j e^{-\beta(\varepsilon_i^A + \varepsilon_j^B)} = z_A z_B$$

N Distinguishable Particles

Similarly,

$$Z = z_A z_B \cdots z_X$$

If the one-particle partition functions are identical ($= z$),

$$Z = z^N$$

Two Indistinguishable Particles

For two indistinguishable particles,

particle 1 in state A & particle 2 in state B

= particle 1 in state B & particle 2 in state A

Hence, the “correct” partition function is

$$Z = \frac{Z_A Z_B}{2}$$

N Indistinguishable Particles

We need to consider Gibbs' correction factor: $1/N!$

$$Z = \frac{1}{N!} z_A z_B \cdots z_X$$

If the one-particle partition functions are identical ($= z$),

$$Z = \frac{1}{N!} z^N$$

Two-State Example

Consider a particle with an internal state: state A or state B.

$$E(A) = \epsilon$$

$$E(B) = -\epsilon$$

The one-particle partition function is

$$z = \sum_s e^{-\beta E(s)} = e^{-\beta E(A)} + e^{-\beta E(B)}$$
$$z = e^{-\beta\epsilon} + e^{+\beta\epsilon} = 2 \cosh(\beta\epsilon)$$

Partition Function

$$z = e^{-\beta\epsilon} + e^{+\beta\epsilon} = 2 \cosh(\beta\epsilon)$$

- Distinguishable particles:

$$Z = z^N = [2 \cosh(\beta\epsilon)]^N$$

- Indistinguishable particles:

$$Z = \frac{1}{N!} z^N = \frac{1}{N!} [2 \cosh(\beta\epsilon)]^N$$

Thermodynamics

- Helmholtz energy

$$A = -k_B T \ln Z = -N k_B T \ln[2 \cosh(\beta \epsilon)]$$

- Internal energy

$$U = -\frac{\partial \ln Z}{\partial \beta} = -N \frac{\partial}{\partial \beta} \ln[2 \cosh(\beta \epsilon)]$$

$$\therefore U = -N \epsilon \tanh(\beta \epsilon)$$

- Entropy: $S = (U - A)/T$

$$S = \frac{1}{T} \{-N \epsilon \tanh(\beta \epsilon) + N k_B T \ln[2 \cosh(\beta \epsilon)]\}$$

